```
onents [201 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Pa
ckages [602 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe Translat
ion-en [72.5 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translat
ion-en [125 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Co
mponents [62.7 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Co
mponents [1056 B]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64
Packages [586 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/restricted Transl
ation-en [122 kB]
Fetched 5008 kB in 8s (610 kB/s)
192 packages can be upgraded. Run 'apt list --upgradable' to see them.
tshark is already the newest version (4.6.4-1).
iperf3 is already the newest version (3.20-2).
mtr-tiny is already the newest version (0.95-1.1ubuntu2).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
wireshark is already the newest version (4.6.4-1).
netcat-openbsd is already the newest version (1.234-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 192
```

```
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2

sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns exec host1 ip a
ddr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up

sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns exec router sys
ctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns exec host1 ping
 -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.078 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.077 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.034 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/mdev = 0.034/0.063/0.078/0.020 ms
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$
```

```
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns exec host1 tsha
rk -i veth-h1 -w /tmp/cap1.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
20 ^C
```

```
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
20 ^C
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$
```

```
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns exec host2 tsha
rk -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
21 ^C
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$
```

```
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
this is mst
three point capture lab

message going thorugh router
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ sudo chmod 644 /tmp/cap1.pcap /tmp/c
ap2.pcap /tmp/cap3.pcap
ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.4K Nov 12 06:37 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.4K Nov 12 06:37 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.4K Nov 12 06:37 /tmp/cap3.pcap
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ tshark -r /tmp/cap1.pcap -T fields -
e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      8 64
      6 63
      8 64
      7 63
      6 64
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ tshark -r /tmp/cap2.pcap -Y "tcp.por
t == 1234" -T fields -e tcp.payload 2>/dev/null | xxd | head -20
00000000: 0a0a 0a34 3836 3536 6336 6336 6632 3036  ...48656c6c6f206
00000010: 3637 3236 6636 6432 3036 3836 6637 3337  6726f6d20686f737
00000020: 3433 3132 3130 610a 0a37 3436 3836 3937  431210a..7468697
00000030: 3332 3036 3937 3332 3036 6437 3337 3430  3206973206d73740
00000040: 610a 0a37 3436 3837 3236 3536 3532 3037  a..7468726565207
00000050: 3036 6636 3936 6537 3432 3036 3336 3137  06f696e742063617
00000060: 3037 3437 3537 3236 3532 3036 6336 3136  074757265206c616
00000070: 3230 610a 3734 3638 3732 3635 3635 3230  20a.746872656520
00000080: 3730 3666 3639 3665 3734 3230 3633 3631  706f696e74206361
00000090: 3730 3734 3735 3732 3635 3230 3663 3631  7074757265206c61
000000a0: 3632 3061 0a0a 3061 0a0a 3664 3635 3733  620a..0a..6d6573
000000b0: 3733 3631 3637 3635 3230 3637 3666 3639  7361676520676f69
000000c0: 3665 3637 3230 3734 3638 3666 3732 3735  6e672074686f7275
000000d0: 3637 3638 3230 3732 3666 3735 3734 3635  676820726f757465
000000e0: 3732 3061 0a0a                           720a..
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$
```

2520090020 CSIT S-11 M.TARUN

```
tarun@LAPTOP-BRD2JBFC:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
this is mst
three point capture lab

message going thorugh router
^C
tarun@LAPTOP-BRD2JBFC:~$
```



